Josephus Problem (9.33)

A. Engel — Problem-Solving Strategies · 1998

Josephus Problem. n persons are arranged in a circle and numbered from 1 to n. Then every kth person is removed and the circle closes up after each removal. What is the number $f(n)$ of the last survivor ?

(a) The problem becomes vastly simplified for $k = 2$. Show that

$$f(2n) = 2f(n) - 1, \qquad f(2n+1) = 2f(n) + 1, \qquad f(1) = 1.$$

Find $f(100)$ by means of these recursions.

(b) There is almost an explicit expression for $f(n)$: Let 2^m be the largest integer, so that $2^m \leq n$. Then

$$f(n) = 2(n - 2^m) + 1.$$

Prove it and find $f(1993)$ by means of this formula.

(c) Write n in the binary system, and transfer the first digit to the end. Then you will get $f(n)$. Show this, and find $f(1000000)$.